Category-Specific Prototype Self-Refinement Contrastive Learning for Few-Shot Hyperspectral Image Classification

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***Abstract*— Deep learning (DL) has been extensively used for hyperspectral image classification (HSIC) with significant success, but the classification of high-dimensional hyperspectral image (HSI) datasets with a limited amount of labeled samples is still a great challenge. Few-shot learning (FSL) has shown excellent performance in solving small-sample classification problems. However, most of the existing FSL methods usually suffer from the prototype instability and domain shift. In order to address these problems, this article proposes a category-specific prototype self-refinement contrastive learning (CPSRCL) method for cross-domain FSL of HSIs. Our method uses a supervised contrastive learning (SCL) strategy to promote intraclass compactness and interclass dispersion of features in the metric space. To stabilize and refine the prototypes of the support set, a category-specific prototype self-refinement (CSPSR) module is designed to adaptively learn different updating rules for different category prototypes using rich labeled information in the query set. Furthermore, a local discriminative domain adaptation (LDDA) method is constructed to align the global distribution between source and target domains while preserving domain-specific discriminative information. Experimental results on four public HSI datasets demonstrate that CPSRCL outperforms the existing FSL and DL methods for HSIC.**

***Index Terms*— Contrastive learning, cross-domain few-shot learning (FSL), hyperspectral image classification (HSIC), prototypical network.**

I. Introduction

HYPERSPECTRAL image (HSI) consists of hundreds of continuous spectral bands with rich geospatial and spectral information. Due to the high sensitivity to spectral differences, HSI has unique advantages for ground object recognition and classification [1], [2]. HSI classification (HSIC) aims to assign each pixel into a land-cover class based on its inherent spectral or spatial characteristics [3]. Over the past decades, HSIC has found extensive applications in various fields, such as ecological environment monitoring, urban planning, and precision agriculture [4], [5].

In the early stages, traditional algorithms that focused on handcrafted features and classifiers were used for HSIC, such as k-nearest neighbor (KNN) [6], support vector machine (SVM) [7], joint sparse representation [8], and Gabor wavelets [9]. Although these algorithms can achieve good classification performance, their handcrafted features cannot handle different scenarios variations adaptively.

In recent years, deep learning (DL) has been extensively used in HSIC and exhibited significant results [10], [11], [12], [13], [14]. In contrast to traditional methods, DL algorithms can automatically extract high-level features of samples through a series of stacked layers and update parameters through backpropagation. Chen et al. [15] proposed a deep HSIC network that extracts rich information from samples with the stacked autoencoder (SAE), revealing the enormous potential of DL networks in HSIC. To reduce the number of parameters, the convolutional neural network (CNN) is designed to HSIC [16] with sparse connectivity and weight sharing strategies. The superior performance of CNN has been demonstrated, and many modified CNN methods have been proposed one after another. In 3-D CNN [17] and hybrid spectral CNN (HybridSN) [18], the convolution structure is modified. In addition, the multiscale structure [19], [20] and attention mechanism [21], [22] have also been applied to improve classification performance. Recently, graph convolutional network (GCN)-based algorithms [23], such as structure preserved hypergraph convolution network and dual-view spectral [24] and global spatial feature fusion network [25], have been proposed for HSIC, which are able to capture the medium- and long-term spatial relationships between samples and greatly improve the classification performance. It is worth noting that the excellent performance of DL-based algorithm is inseparable from a large number of labeled samples. Unfortunately, labeling samples is time-consuming and laborious, and the number of available labeled samples is extremely small.

Several strategies have been introduced to solve the label scarcity problem in DL-based HSIC. For example, data augmentation strategies, such as random Gaussian noise, random

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rotation, and cropping [26], [27], can be used to augment the number of labeled samples by performing transformations on the available samples. Lightweight networks are designed to reduce network parameters, and the reliance on labeled samples can also be reduced. Typical lightweight networks usually use depthwise convolution [28] or pointwise convolution [29]. In addition, self-supervised learning can be used to mine the implicit supervised information of unlabeled samples and shows good results for small-sample classification [30], [31], [32] or other pretext tasks [33].

During the past several years, few-shot learning (FSL) has emerged as a powerful technique for identifying new unseen categories with limited labeled samples. Unlike small-sample classification, FSL can train with labeled samples from a known dataset and classify unlabeled samples on a new dataset. Recently, FSL has also been applied for small-sample HSIC [34]. For instance, Liu et al. [35] designed a deep FSL (DFSL) method for classifying HSI using limited labeled data, which incorporates the episode training paradigm for HSIC and learns the experiences or patterns of classification in a randomly changing meta-task. Similarly, Gao et al. [36] proposed a deep relation network for few-shot HSIC (RN-FSC), which makes the trained model more applicable to the test dataset by fine-tuning operations. Furthermore, Li et al. [37] proposed a deep cross-domain FSL (DCFSL) method to investigate the domain shift problem in the FSL. DCFSL uses an adversarial domain-based adaptation module to mitigate the domain bias phenomenon. Xi et al. [38] introduced a class-covariance FSL (CMFSL) method that designs the refinement module to reduce interdomain differences in conjunction with a class-covariance-based metric for classification. Zuo et al. [39] proposed a graph neural network based on edge labeling for few-shot HSIC (FSL-EGNN). FSL-EGNN embeds edge-labeling graph neural network (EGNN) in the FSL framework and achieves accurate classification of unknown samples by cyclically updating the node features and edge features. Zhang et al. [40] proposed a graph information aggregation network for cross-domain FSL (Gia-CFSL), which uses graph information aggregation strategy to suppress domain shift phenomenon.

A challenging problem in FSL-based methods is the domain difference between the source and target domains. The existing FSL-based HSIC methods use domain adaptation (DA) strategies [41], [42], [43], [44], [45] directly to address possible domain shifts. However, FSL differs significantly from DA, where the feature space and label space between domains are the same. In contrast, FSL can handle entirely different classes and features in two domains. Traditional DA methods may not be effective in the FSL tasks. Fig. 1(a) shows that these traditional DA methods aim to align distribution of two domains. Even if global distribution between domains is aligned, it often leads to confusion between categories, which reduces the discriminative ability in FSL. Another problem in FSL methods is the inaccuracy and instability in the calculation of the prototypes of support set. In the original metric-based FSL method, the prototype of each category is computed by directly averaging sample features in the corresponding category. Obviously, noisy samples will affect the calculation of prototypes. To improve prototype calculation, a number of prototype-based FSL strategies have been developed [46], [47], [48]. For instance, Cheng et al. [49] proposed a Siamese-prototype network that refines support set prototypes by two calibration modules. Yu et al. [50] proposed a multiview calibrated prototype learning (MCPL) method for few-shot HSIC, which converts prototypes from the feature mean vector to local HSI patch features via a calibrated aggregation network (CAN). Liu et al. [51] developed a refined prototypical network based on contrastive learning (RPCL) method that leverages contrastive learning to impose three restrictions on the support set prototypes. Xu et al. [52] proposed a prototype self-updating (PSU) method for FSL that uses a PSU module to improve the prototype. However, as shown in Fig. 1(b), these methods optimize all category prototypes uniformly, which may cause some prototypes to deviate from their real versions.

To solve the aforementioned domain shift and prototype instability problems, we propose a category-specific prototype self-refinement contrastive learning (CPSRCL) model for cross-domain few-shot HSIC. In the proposed CPSRCL, as illustrated in Fig. 1, a supervised contrastive learning (SCL) strategy is used to highlight interclass variability of features, a local discriminative DA (LDDA) strategy is designed to tackle the domain shift problem, and a category-specific prototype self-refinement (CSPSR) subnetwork is constructed to update and calibrate support set prototypes adaptively. The main contributions of our proposed CPSRCL are as follows.

1) We design a CSPSR module to refine the prototypes of support set using categories aggregation information. It can generate category-specific feature mapping to guide the updating of prototype of each category, which can further enhance the stability and effectiveness of support set prototypes.
2) We introduce an improved DA strategy LDDA to mitigate the domain shift problem in cross-domain FSL, which can align global distribution between source and target domains and, meanwhile, maintain domain-specific discriminative information.
3) We incorporate the SCL into the FSL framework by constructing the contrastive support set to further emphasize intraclass similarity and interclass variability of features.

The article consists of the following sections. Section II introduces the related works. Section III describes the CPSRCL method. The experimental results are presented in Section IV. Finally, Section V draws a conclusion.

II. Related Work

A. Cross-Domain FSL

FSL is an application of meta-learning, which intends to allow machines to learn to "learn" like humans and to perform the similar types of tasks with prior knowledge [51]. FSL methods mainly have three groups depending on the training strategy. The first category is the model-based learning strategy [53], [54], which fine-tunes the trained model from the source domain categories and then quickly adapts to the target domain classes. The second group is the metric-based learning method

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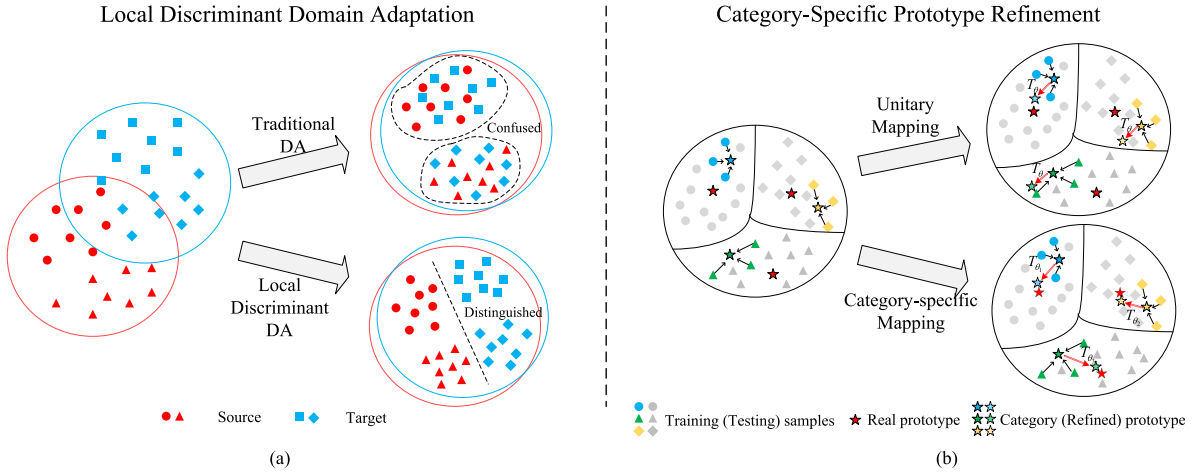


Fig. 1. Illustration of DA and prototype refinement modules. (a) Large red and blue circles represent the global distribution of source and target domains, respectively, and the small markers represent four different categories in two domains (the categories of two domains in FSL can be different). Traditional DA methods align the global distribution between domains (e.g., the red and blue circles almost coincide) while also force to align different categories in two domains. The local discriminant DA method attempts to align global distributions while preserving category-specific distribution and discriminative information in both domains. (b) Original metric-based FSL method directly uses the averaged features as category prototypes, and the obtained prototypes usually deviate from real prototypes, as shown in the left part. The existing prototype refinement methods adopt unitary mapping to update all prototypes, which can only improve part prototypes. The category-specific prototype refinement strategy develops different updating mappings for different prototypes. The refined prototypes are more closer to real prototypes.

[55], [56], [57], which mainly learns a similarity metric, such that similar samples have high similarity scores. The third group is the optimization-based strategy [58], which learns updating functions or rules for the model parameters to fit in the few-shot regime. Among the above approaches, intuitive and effective metric-based strategies are the most widely used in the HSIC field.

The training process of FSL involves two datasets: a training dataset with sufficient labeled samples and a testing dataset with only a small number of labeled samples. The training and testing datasets may come from different domains with severe domain shifts. For HSI-based FSL, the training and testing datasets may acquired by different sensors at different imaging environments, e.g., Chikusei and Indian Pines (IP) datasets. This problem is called cross-domain FSL, where the training source domain sample set is denoted as D_{sd} with C_{sd} classes and the testing target domain sample set is denoted as D_{td} with C_{td} classes. In general, in order to obtain more information about the training dataset, C_{sd} should be no less than C_{td}. Since the categories in the two domains are usually different, traditional DA strategies may not be applicable to cross-domain FSL.

B. Prototypical Networks

ProtoNet [59] learns a metric space by computing the distance of the testing sample from the prototype of each category. In the FSL setting, the paradigm of episode training was used to mimic the few-shot task [37]. Taking the source domain as an example, for each episode task, N categories are randomly selected from C_{sd} categories; then, K labeled samples are sampled from each of the selected N categories to compose the support set, and then, several samples per class are selected from the remaining samples to form the query set, which is the N-way K-shot problem. The model is able to extract the common characteristics of different meta-tasks and, thus, classify new unseen tasks efficiently. However, ProtoNet calculates the mean of features for each category as a category prototype, and the noisy samples may severely affect the quality of support set prototypes, which will lead to unstable classification performance.

C. Supervised Contrastive Learning

Contrastive learning [60] is a widely used self-supervised learning strategy to improve the identification ability with the idea of pulling positive samples together and pushing negative samples far away. Khosla et al. [61] proposed fully SCL based on self-supervised contrastive learning (SSCL), which uses the sample label information to keep samples of the same category closer to each other and samples of different categories farther apart, i.e.,

$$\text{distance}(f(x), f(x^+)) \gg \text{distance}(f(x), f(x^-)) \quad (1)$$

where distance$(\cdot)$ denotes the degree of resemblance of the two values, x and x^+ belong to the same category, and x and x^- are from different categories. The support set of FSL consists of $N \times K$ labeled samples. When $K = 2$, i.e., two labeled samples per category are selected, for each sample x in the support set, x^+ represents the only sample from the same category, and x^- represents the $2N - 2$ samples of different categories.

III. Methodology

In this section, we describe our CPSRCL method for cross-domain FSL in detail. First, we introduce the problem setup about cross-domain FSL in Section III-A. Then, the overall framework of CPSRCL is shown in Section III-B. Sections III-C–III-F detail four main modules of CPSRCL, i.e., data processing, LDDA, SCL, and CSPSR.

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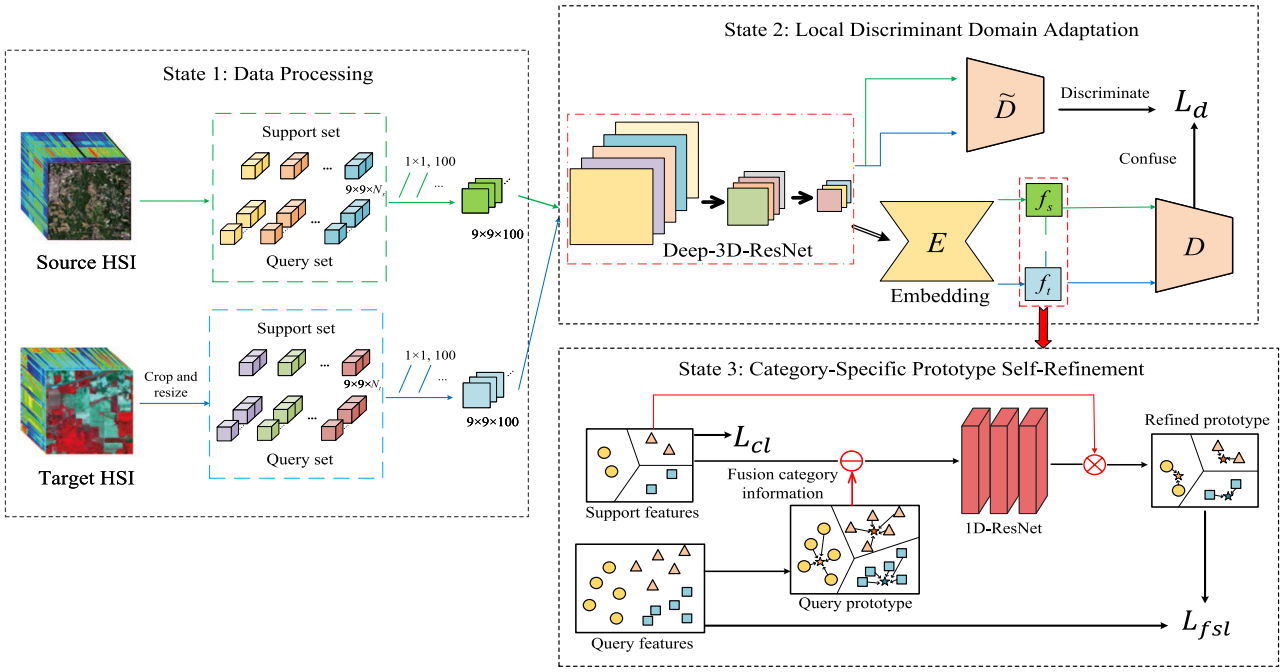


Fig. 2. Framework of CPSRCL. The LDDA strategy mitigates the domain shift between two domains by imposing domain discrimination and confusion. Samples from two domains are alternately trained for the FSL task, and CSPSR strategy acting on f_s and f_t after the embedding layer is used to obtain more stable prototypes.

A. Problem Setup

The objective of meta-learning is to allow machines to learn generalized knowledge as categories change, so that they can classify new unseen categories without changing the existing models. FSL is an application of meta-learning to the field of supervised learning, which identifies new categories with only a small number of additional labeled samples. The ProtoNet-based FSL method is widely used for small-sample HSIC. Concretely, ProtoNet requires learning a metric space with different meta-tasks, in which the machine learns the common parts of different tasks and ignores the task-specific parts. For each meta-task, the training set in an episode task is usually described as an N-way K-shot problem.

For cross-domain FSL, the two datasets involved are the source domain dataset D_{sd} with C_{sd} classes and the target domain dataset D_{td} with C_{td} classes, where the two datasets have completely different categories and C_{td} cannot be larger than C_{sd}. In this article, we set a C_{td}-way two-shot training paradigm for all datasets (i.e., $N = C_{\text{td}}$ and $K = 2$), which means that the support set S_{sd} in each task contains C_{td} categories and each category has only two labeled samples. In addition, as in [35] and [37], we randomly select 19 labeled samples from each category of the remaining samples to form the query set Q_{sd}.

For the traditional ProtoNet-based FSL methods, the category prototype is obtained by directly calculating the sample mean for each category. The application of ProtoNet-based FSL for small-sample HSIC may suffer from the following problems: 1) due to the limited number of labeled samples in each class (e.g., 2), each sample plays an important role when computing the category prototype, and the noisy or less informative samples will affect the prototype and 2) the dataset used for training and the testing dataset are from completely different domains, and there is a large interdomain difference between the two datasets.

B. Overall Framework

Considering the above problems, we propose a CPSRCL model for cross-domain few-shot HSIC. The overall framework of CPSRCL is shown in Fig. 2, where the CSPSR module allows the prototypes of support set to be self-updated continuously, as the network iterates to solve the prototype instability problem, and the LDDA strategy can mitigate domain shifts.

For each training process, there are three stages. First, data augmentation strategy and adaptive mapping layers are used to ensure the same spectral dimension and the same number of labeled samples between two domains. Second, few-shot DA strategy is used to mitigate the differences between two domains. Finally, a category-specific mapping acting on the support set is designed for the refinement of the prototypes, which allows the prototypes to be updated adaptively during network iterations. In addition, a contrastive learning strategy is used to enhance the performance of the feature extractor.

The meta-testing phase is shown in Fig. 3. The labeled and unlabeled samples in the target dataset are sequentially fed into the mapping layer, the deep 3-D ResNet layer, and the embedding layer to generate domain-invariant embedding features. Finally, the 1-nearest neighbor (NN) classifier is used to classify unlabeled samples.

C. Data Processing

Considering that the target domain has a very limited number of labeled samples for FSL training, a data

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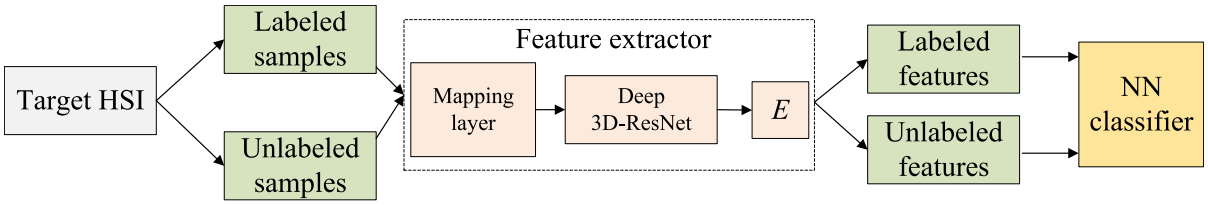


Fig. 3. Flowchart of classification in the testing phase.

augmentation strategy (random cropping and resizing) [51] is used to expand the training set of the target domain. After repeating the above operation several times, there are 200 labeled samples per class in the training set of the target domain. We build support and query sets on the source and target domains; taking the source dataset as an example, N categories are randomly selected from the C_{sd} training classes; then, two labeled samples are randomly sampled from each class to form the support set $S_{\text{sd}} = \{(x_i, y_i)\}_{i=1}^{n_s}$, and 19 labeled samples [35], [37] are selected to form the query set $Q_{\text{sd}} = \{(x_j, y_j)\}_{j=1}^{n_q}$, where $N = C_{\text{td}}$, $n_s = 2 \times N$, and $n_q = 19 \times N$. Inspired by the article [51], we divide the labeled samples of the support set into two separate groups, each containing samples of different classes, and then obtain the contrastive support set, which facilitates the application of subsequent SCL.

Due to the differences in imaging sensors, the spectral dimensions of the source and target domain data are usually different. An adaptive mapping layer [37] is used to unify the spectral dimension. The adaptive mapping layer uses a simply convolutional operation through c convolution kernels of size $1 \times 1 \times h$, where h is the dimension of the input and c is the dimension of the output ($c = 100$ in the experiments).

D. DA for FSL

The deep 3-D residual network [35] with an embedding module is used as the feature extractor. As shown in Fig. 4, the feature extraction network consists of two residual blocks and a embedding block, where each residual block contains three 3-D convolutional layers with eight convolution kernels of size $3 \times 3 \times 3$, and the embedding block consists of two linear layers. The intermediate and final features of the input samples are extracted before and after the embedding block, respectively.

Considering that the source and target domains in FSL have different categories, traditional DA strategies may be unsuitable for cross-domain FSL problems. Inspired by Zhao et al. [62], we aim to align the global distribution of the source and target domains, while part of the domain-specific information is still preserved. To this end, we implement domain discrimination and domain alignment on intermediate and final features, respectively.

1) Domain Alignment Loss: Motivated by the paper [63], we use a conditional domain discriminator D across the source distribution $P_{\text{sd}}(x)$ and the target distribution $P_{\text{td}}(x)$ after the embedding layer to reduce domain shift. Denoting the final feature representation as $f = F(x)$ and the classifier prediction as $g = G(x)$ after embedding module, the loss

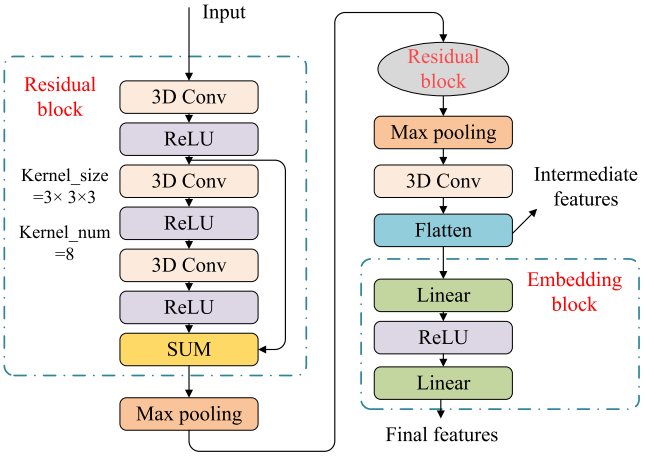


Fig. 4. Architecture of the feature extractor. The network consists of two residual blocks and one embedding block. The intermediate and final features are extracted before and after the embedding block, respectively.

function L on the discriminator D can be defined as follows:

$$\min_{D} \max_{F,G} L = -E_{x_i \sim P_{\text{sd}}(x)} \log \left[D\left(f_i^{\text{sd}}, g_i^{\text{sd}}\right) \right] - E_{x_j \sim P_{\text{td}}(x)} \log \left[1 - D\left(f_j^{\text{td}}, g_j^{\text{td}}\right) \right] \quad (2)$$

where $D(,)$ represents the probability that the sample x belongs to the source domain.

In order to fully capture the multimodal structures behind complex data distributions, the multilinear map $T_{\otimes}(h) = f \otimes g$ is used in the operation instead of the traditional concatenation strategy. To facilitate the computation, the inner product $T_{\otimes}(h)$ can be approximated by the dot product $T_{\odot}(h) = (1/\sqrt{d})(R_f f)(R_{gg})$, where $R_f f$ and R_{gg} are two random matrices that are sampled only once and fixed in the training phase, and $d \ll d_f \times d_g$ (d_f and d_g are the dimensions of f and g, respectively). Finally, the conditioning strategy was adapted as follows:

$$T(h) = \begin{cases} T_{\otimes}(h), & d_f \times d_g \leq d_{\text{fea}} \\ T_{\odot}(h), & \text{otherwise} \end{cases} \quad (3)$$

where d_{fea} represents the output dimension of the network, which is set to 1024 in the experiments. This means that the strategy of randomized multilinear map $T_{\odot}$ is chosen when the dimensions of multilinear map $T_{\otimes}$ exceed 1024.

The domain alignment loss ℓ_{da} in (2) can be rewritten as follows:

$$\ell_{\text{da}} = -E_{x_i \sim P_{\text{sd}}(x)} \log \left[D\left(T\left(h_i^{\text{sd}}\right)\right) \right] - E_{x_j \sim P_{\text{td}}(x)} \log \left[1 - D\left(T\left(h_j^{\text{td}}\right)\right) \right]. \quad (4)$$

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The domain discriminator D [37] used to perform the domain alignment is a multilayer perceptron (MLP), which consists of five fully connected layers and four rectified linear units (ReLUs).

2) Domain Discriminative Loss: When the source and target domains share the same categories, the above domain alignment strategy is effective. However, for cross-domain FSL, two domains have different categories, and excessive domain alignment operations may cause the model to lose discriminative ability over new categories.

To preserve domain-specific information, a local domain discriminant method is used to restrict intermediate features before the embedding layer. A discriminator $\widetilde{D}$ is used across the source distribution $P_{\text{sd}}(x)$ and the target distribution $P_{\text{td}}(x)$ before the embedding layer. Here, we want the distributions of categories in two domains to be different from each other. Similarly, we can define a domain discriminative loss $\widetilde{L}$ on the discriminator $\widetilde{D}$ as follows:

$$\min_{\widetilde{D},\widetilde{F},\widetilde{G}} \widetilde{L} = -E_{x_i\sim P_{\text{sd}}(x)} \log\left[\widetilde{D}\left(\widetilde{f}_i^{\text{sd}}, \widetilde{g}_i^{\text{sd}}\right)\right] - E_{x_j\sim P_{\text{td}}(x)} \log\left[1-\widetilde{D}\left(\widetilde{f}_j^{\text{td}}, \widetilde{g}_j^{\text{td}}\right)\right]. \tag{5}$$

Let $\widetilde{h} = (\widetilde{f}, \widetilde{g})$, and the domain discriminative loss ℓ_{dd} is defined as follows:

$$\ell_{\text{dd}} = -E_{x_i\sim P_{\text{sd}}(x)} \log\left[\widetilde{D}\left(T\left(\widetilde{h}_i^{\text{sd}}\right)\right)\right] - E_{x_j\sim P_{\text{td}}(x)} \log\left[1-\widetilde{D}\left(T\left(\widetilde{h}_j^{\text{td}}\right)\right)\right]. \tag{6}$$

It is noteworthy that we use the domain classifier $\widetilde{G}$ as the domain discriminator $\widetilde{D}$ and directly calculate the classification loss for different domain samples. The DA loss L_d can eventually be expressed as follows:

$$L_d = \ell_{\text{da}} + \ell_{\text{dd}}. \tag{7}$$

E. SCL Strategy

The ProtoNet-based FSL is trained by minimizing the distance between prototypes and samples of the same category, which enables samples of the same category to cluster more tightly in the new embedding space. However, the ProtoNet-based FSL only considers the intraclass similarity and ignores the interclass variability, which may lead to confusion of similar categories. To alleviate this problem, we employ SCL to enhance the discriminative performance of the model.

SCL serves as a generalization of SSCL, which has better results in specific scenarios, such as sample sets with labeled information. Unlike SSCL, SCL utilizes the label of samples to artificially set samples of the same category as positive samples, while all other samples of different categories are considered as negative samples. Thus, it is possible to maximize the assurance that all negative samples come from different categories. Moreover, whereas the core of SSCL is to have enough negative pairs to contrast with the positive pairs, SCL not only retains this property, but also increases the number of positive pairs.

For each N-way two-shot meta-task, each category in the support set contains two labeled samples, and two individual contrastive groups can be constructed, where each group consists of samples from different categories. We denote these two groups as $S_1 = \{x_1, x_3, x_5, \ldots, x_{2N-1}\}$ and $S_2 = \{x_2, x_4, x_6, \ldots, x_{2N}\}$, respectively, where x_{2k-1} and x_{2k} are from the same category. The contrastive loss function $\ell_{m,n}$ between $F(x_m)$ and $F(x_n)$ can be defined as follows:

$$\ell_{m,n} = -\log \frac{\exp\left(\frac{s(F(x_m),F(x_n))}{\tau}\right)}{\sum_{k=1}^{2N} \mathbf{1}_{[k\neq m]} \exp\left(\frac{s(F(x_m),F(x_k))}{\tau}\right)} \tag{8}$$

where $x_m, x_n \in S_{\text{sd}}$ belong to the same category and are mutually positive samples, $s(,)$ represents the Euclidean-distance-based similarity metric, and τ is a temperature coefficient, which is set to 0.5 in our method.

Considering that $\ell_{2k-1,2k} \neq \ell_{2k,2k-1}$, the final average contrastive loss L_{cl} for each meta-task is

$$L_{\text{cl}} = \frac{1}{2N}\sum_{k=1}^{N}(\ell_{2k-1,2k} + \ell_{2k,2k-1}). \tag{9}$$

F. Category-Specific Prototype Self-Refinement Strategy

The prototype of each category in the ProtoNet-based FSL guides the query samples to complete the classification, and its quality greatly impacts on the model performance. However, in the case of limited labeled samples, the support set prototypes obtained directly from the features averaging are prone to be affected by noisy samples. The existing prototype refinement methods constantly update the category prototypes by parameter restriction or construction of common subspaces, ultimately resulting in more representative prototypes. However, these methods use a single unified update strategy for different classes of prototypes, which only improves part prototypes. To improve the robustness of prototypes, a CSPSR strategy is proposed. Specifically, we make full use of the label information in the query set by aggregating the query set information and support set samples from the same category before feeding the support set samples into the updating network. As a result, the network can focus on different categories and enhance corresponding prototypes. It allows the prototypes to be continuously updated with the iterations of the network.

The framework of the CSPSR module is shown in Fig. 5. First, we combine the query set prototypes with the sample features of the support set. The combined features are then fed into the embedding network $T_\theta(\cdot)$ to extract category-specific aggregated information. Finally, the aggregated information is assigned to the original support set features to update support set prototypes.

The prototype of class c in the query set is obtained by calculating the mean value

$$P_q^c = \frac{1}{n_q^c}\sum_{x_j\in Q_{\text{sd}}^c} F(x_j) \tag{10}$$

where Q_{sd}^c represents the query set samples belonging to class c and n_q^c is the number of samples per class.

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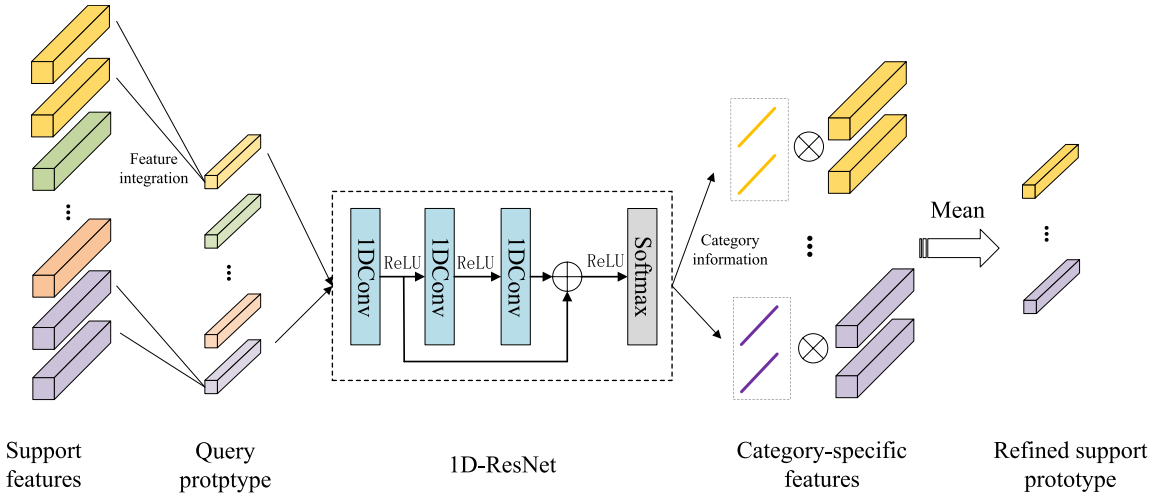


Fig. 5. Illustration of category-specific prototype refinement module. First, the category prototypes of the support and query sets are computed from the mean of the labeled samples; then, their information is aggregated and fed into a 1-D residual network to extract the category information, and finally, the category information is fused with the original support set samples to obtain the refined support prototypes.

We can construct a category-specific mapping $T_{\theta_n}(\cdot)$ with respect to the query set category prototypes and add it to the network training, which is capable to adaptively transform samples from different classes. The aggregated information ω_i for the support set feature $F(x_i)$ can be calculated as follows:

$$\omega_i = T_{\theta_{y_i}}(F(x_i)) \tag{11}$$

where y_i is the label of x_i (belonging to $1, \ldots, N$) and

$$T_{\theta_n}\big(F(x_i)\big) = T_\theta\big(P_q^n - F(x_i)\big), \quad n = 1, \ldots, N. \tag{12}$$

The universal mapping $T_\theta(\cdot)$ is restricted to be category-specific, and each category has a unique mapping $T_{\theta_n}(\cdot)$.

The aggregated support set features can be obtained as follows:

$$F_{\text{new}}(x_i) = \sum_{i=1}^{n_s} F(x_i) \cdot \omega_i. \tag{13}$$

Then, the refined prototype of class c in the support set can be obtained as follows:

$$P_s^c = \frac{1}{n_s^c} \sum_{x_i \in S_{\text{sd}}^c} F_{\text{new}}(x_i) \tag{14}$$

where S_{sd}^c represents the support set samples belonging to class c and n_s^c is the number of samples per class.

Finally, the FSL loss L_{fsl} for each meta-task is

$$L_{\text{fsl}} = -\frac{1}{n_q} \sum_{j=1}^{n_q} \log \frac{\exp\big(-\text{dist}\big(F(x_j), P_s^{y_j}\big)\big)}{\sum_{c=1}^{N} \exp\big(-\text{dist}\big(F(x_j), P_s^c\big)\big)} \tag{15}$$

where $\text{dist}(\cdot)$ represents the Euclidean distance of the two features.

Finally, the total loss function of each training process of CPSRCL is

$$L_{\text{total}} = L_{\text{fsl}} + L_d + L_{\text{cl}}. \tag{16}$$

The detail of the training procedure of the CPSRCL method is shown in Algorithm 1.

Algorithm 1 Training Process of CPSRCL

Input: The intermediate feature $\widetilde{F}(x)$, the final feature $F(x)$, S_{sd} and Q_{sd} in D_{sd}, S_{td} and Q_{td} in D_{td}
Output: L_{total}

```
if epoch % 2 = 0 then
    Train with S_sd and Q_sd
    Calculate the prototype of query set Q_sd → P_q
    for x_m, x_n ∈ S_sd do
        Calculate the L_cl by Eq. (9)
    end for
    for x_i ∈ S_sd do
        Calculate the category information for each support set sample by Eq. (11) and (13)
        Calculate the refined prototype by Eq. (14)
        Calculate the L_fsl by Eq. (15)
    end for
    for Intermediate feature F̃(x) do
        Calculate the domain alignment loss ℓ_da by Eq. (4)
    end for
    for Final feature F(x) do
        Calculate the domain discriminative loss ℓ_dd by Eq. (6)
    end for
    Return the loss L_total by Eq. (16)
else
    Train with S_td and Q_td alternately
end if
```

IV. Experiment

A. Datasets

Five public datasets are used in the experiments, including the Chikusei, University of Pavia (UP), IP, Salinas (SA), and HuangheKou (HHK). In the FSL setting, the Chikusei is taken as the source domain, and UP, IP, SA, and HHK are selected as target domain.

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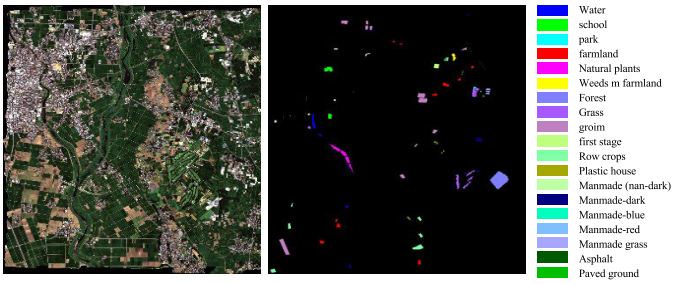


Fig. 6. False-color image and ground-truth map of Chikuse.

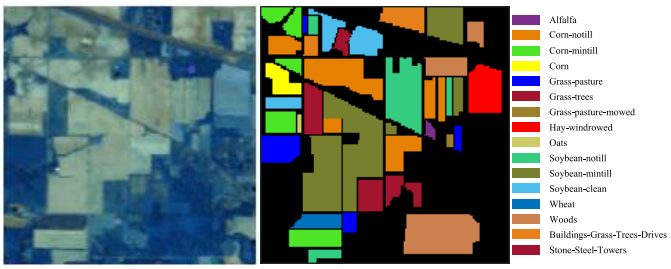


Fig. 7. False-color image and ground-truth map of IP.

TABLE I
CATEGORIES AND NUMBERS OF SAMPLES IN CHIKUSEI

Class	Name	Pixels
1	Water	2345
2	school	2859
3	park	236
4	farmland	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	groim	13369
10	first stage	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (nan-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	Total	77592

TABLE II
CATEGORIES AND NUMBERS OF SAMPLES IN IP

Class	Name	Pixels
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
Total		10249

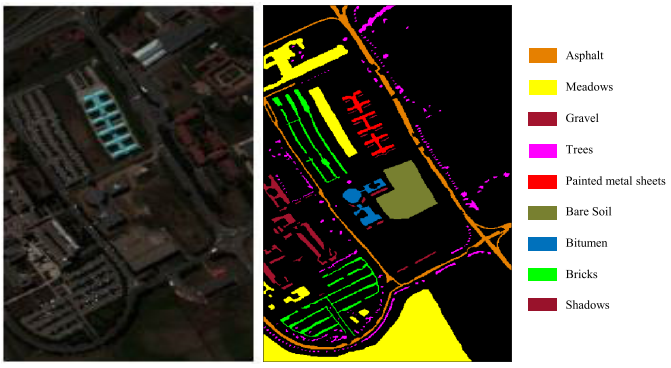


Fig. 8. False-color image and ground-truth map of UP.

The source and target datasets are as follows.

1) Source Domain Dataset: The Chikusei dataset contains 128 spectral bands ranging from 343 to 1018 nm, with a spatial size of 2517 × 2335 and a spatial resolution of 2.5 m. It has 19 land-cover classes that encompass both urban and rural areas. The details of the dataset are shown in Fig. 6 and Table I.

2) Target Domain Datasets:

1) The IP dataset contains 200 spectral bands ranging from 400 to 2500 nm, with a spatial size of 145 × 145 and a spatial resolution of 20 m. It has 16 land-cover classes. The details of the dataset are shown in Fig. 7 and Table II.
2) The UP dataset contains 103 spectral bands ranging from 430 to 860 nm, with a spatial size of 610 × 340 and a spatial resolution of 1.3 m. It has nine different types of land-cover classes. The details of the dataset are shown in Fig. 8 and Table III.
3) The SA dataset contains 224 spectral bands ranging from 400 to 2500 nm, with a spatial size of 512 × 217 and a spatial resolution of 20 m. It has 16 different types of vegetation categories. The details of the dataset are shown in Fig. 9 and Table IV.
4) The HHK dataset contains 285 spectral bands ranging from 390 to 2513 nm, with a spatial size of 1185 × 1342 and a spatial resolution of 30 m. It has 21 different types of categories. To satisfy the FSL condition ($C_{td} \leq C_{sd}$), we have removed categories 20 and 21 (Tamarix and Intertidal Saltwater) and fused categories 6 and 8 into a new category (Sea). The details of the dataset are shown in Fig. 10 and Table V.

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TABLE III
CATEGORIES AND NUMBERS OF SAMPLES IN UP

Class	Name	Pixels
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Metal sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadows	947
Total		42776

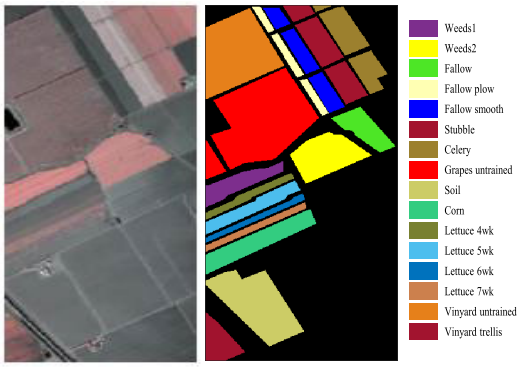


Fig. 9. False-color image and ground-truth map of SA.

TABLE IV
CATEGORIES AND NUMBERS OF SAMPLES IN SA

Class	Name	Pixels
1	Brocoli_green_weeds 1	2009
2	Brocoli_green_weeds 2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
Total		54129

3) Experiment Setting: We compare the CPSRCL with several DL- and FSL-based algorithms, including HybirdSN [18], attention-based adaptive spectral-spatial kernel ResNet (A2S2K) [64], spectral–spatial residual network (SSRN) [65], SSCL [66], DFSL [35], DCFSL [37], CMFSL [38], and Gia-CFSL [40]. The first four algorithms are DL-based

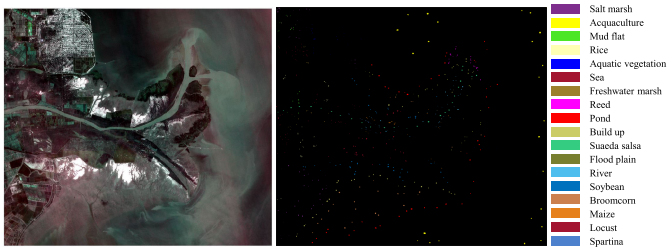


Fig. 10. False-color image and ground-truth map of HHK.

TABLE V
CATEGORIES AND NUMBERS OF SAMPLES IN HHK

Class	Name	Pixels
1	Salt marsh	393
2	Acquaculture	796
3	Mud flat	110
4	Rice	190
5	Aquatic vegetation	83
6	Sea	307
7	Freshwater marsh	95
8	Reed	200
9	Pond	936
10	Build up	553
11	Suaeda salsa	469
12	Flood plain	361
13	River	240
14	Soybean	595
15	Broomcorn	454
16	Maize	133
17	Locust	377
18	Spartina	68
Total		6360

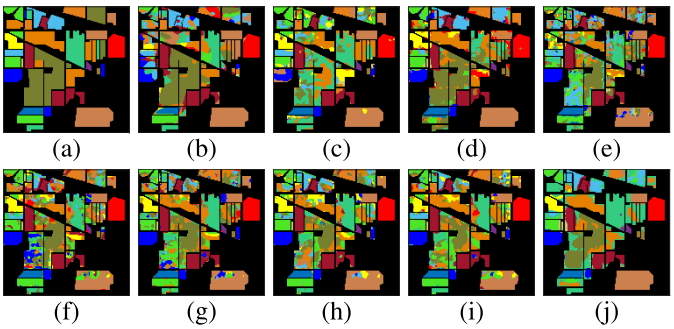


Fig. 11. Classification map of different algorithms on the IP dataset. (a) Ground truth. (b) HybirdSN (62.16%). (c) SSRN (61.28%). (d) A2S2K (69.76%). (e) SSCL (56.81%). (f) DFSL (60.12%). (g) DCFSL (64.87%). (h) CMFSL (67.07%). (i) Gia-CFSL (66.27%). (j) CPSRCL (77.65%).

classification methods, which do not involve the source and target domains. These four methods are directly performed on the target domain dataset, where five labeled samples per class are randomly selected to train the network. The remaining four algorithms are FSL-based methods, among which DFSL and CMFSL fuse four datasets (i.e., Chikusei, Houston 2013, Botswana, and Kennedy Space Center) as the source domain dataset and unify the dimensions of two domains using the band selection strategy proposed in [67], and DCFSL and

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TABLE VI
EXPERIMENTAL RESULTS (%) OF DIFFERENT ALGORITHMS ON IP

Class	HybirdSN	SSRN	A2S2K	SSCL	DFSL	DCFSL	CMFSL	Gia-CFSL	CPSRCL
1	89.39	80.28	36.19	87.3	80.24	90.17	94.67	**100**	98.78
2	63.82	44.18	68.28	36.82	30.17	54.23	49.21	66.37	**71.05**
3	61.27	48.38	65.34	54.33	79.24	72.88	**84.19**	80.13	64.06
4	42.85	50.52	49.25	34.61	87.17	83.72	**93.27**	75.18	92.03
5	50.31	92.37	87.68	73.17	85.33	**93.27**	92.31	91.08	86.19
6	78.29	96.51	97.21	89.95	88.48	92.68	**97.35**	96.28	86.90
7	78.58	47.29	20.73	**100**	**100**	**100**	**100**	**100**	**100**
8	91.11	99.75	47.85	97.72	**100**	98.85	99.11	99.76	97.67
9	27.28	28.42	25.18	**100**	**100**	**100**	**100**	**100**	**100**
10	55.51	45.28	64.29	49.26	54.81	67.24	67.28	68.38	**76.01**
11	71.32	71.73	74.39	34.27	42.32	40.38	38.13	37.18	**73.20**
12	45.73	45.73	56.27	43.37	37.28	30.60	48.92	49.38	**56.80**
13	90.55	64.64	76.19	98.48	97.66	97.47	96.18	**98.57**	96.25
14	69.17	**95.82**	92.27	73.28	73.12	83.15	84.37	72.39	93.21
15	31.28	21.38	51.11	65.58	58.37	66.52	64.57	**81.34**	64.04
16	25.62	24.41	57.53	96.18	**100**	**100**	99.19	**100**	99.43
OA	62.16	61.28	69.76	56.81	60.12	64.87	67.07	66.27	**77.65**
AA	60.76	59.79	60.64	70.9	75.89	79.45	81.80	82.25	**84.73**
Kappa	61.37	56.87	66.84	49.85	55.91	60.95	62.83	63.19	**74.63**

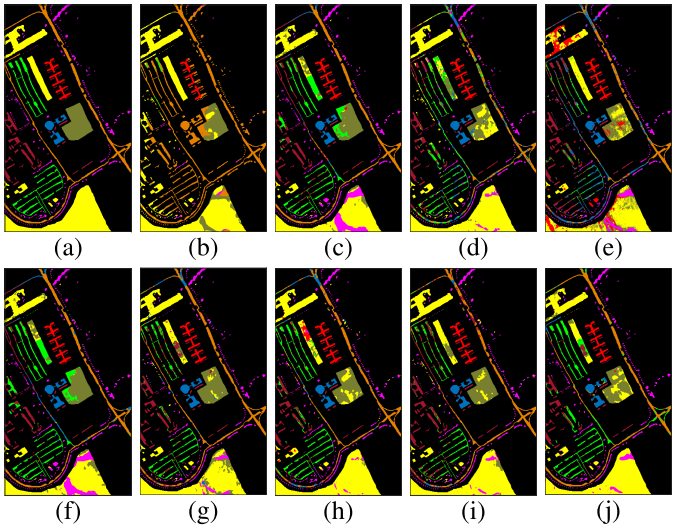


Fig. 12. Classification map of different algorithms on the UP dataset. (a) Ground truth. (b) HybirdSN (74.11%). (c) SSRN (79.81%). (d) A2S2K (68.43%). (e) SSCL (62.28%). (f) DFSL (78.24%). (g) DCFSL (82.24%). (h) CMFSL (85.38%). (i) Gia-CFSL (83.72%). (j) CPSRCL (87.74%).

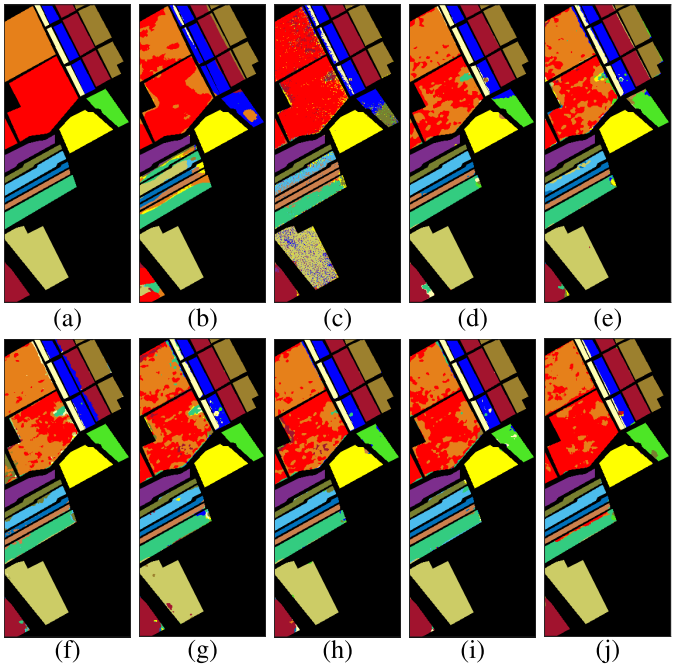


Fig. 13. Classification map of different algorithms on the SA dataset. (a) Ground truth. (b) HybirdSN (83.82%). (c) SSRN (77.68%). (d) A2S2K (88.27%). (e) SSCL (86.32%). (f) DFSL (84.88%). (g) DCFSL (86.83%). (h) CMFSL (89.45%). (i) Gia-CFSL (89.24%). (j) CPSRCL (91.64%).

Gia-CFSL use the same source domain dataset (i.e., Chikusei) and the same dimensionality mapping method as in this article.

In our CPSRCL, 9×9 neighborhoods are the size of input cube, the learning rate is 0.001, and the number of training iterations is set to 3000. In an N-way K-shot episode training, N equals the number of target domain dataset categories (e.g., $N = 9$ for UP and $N = 18$ for HHK), and K is set to 2. In addition, we select 19 labeled samples from each category of the remaining samples to form the query set. The classification performance is evaluated using the overall accuracy (OA), average accuracy (AA), and k coefficient. The OA refers to the number of all correctly categorized samples divided by the total number of samples. AA refers to the average of the categorization accuracies for each category. Kappa coefficients are used in consistency tests to measure whether the model predictions are consistent with the actual classification results. The averaged results in ten randomly runs are reported.

B. Classification Results

The experimental results of the IP dataset are shown in Table VI. It can be seen that our algorithm is about 10% higher than other algorithms in terms of OA and k coefficient. For categories 2, 7, and 9–12, the proposed CPSRCL has the

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TABLE VII
EXPERIMENTAL RESULTS (%) OF DIFFERENT ALGORITHMS ON UP

Class	HybirdSN	SSRN	A2S2K	SSCL	DFSL	DCFSL	CMFSL	Gia-CFSL	CPSRCL
1	60.42	**99.58**	92.38	56.41	98.43	83.24	71.24	74.53	86.91
2	81.34	97.68	84.24	64.13	**98.64**	80.14	92.21	91.15	88.94
3	34.14	49.37	56.24	79.48	48.22	90.38	80.81	**89.28**	80.06
4	92.48	51.26	69.38	73.23	50.23	88.22	89.37	84.44	**93.30**
5	**100**	**100**	96.41	**100**	99.42	**100**	**100**	**100**	99.40
6	92.37	91.31	57.95	62.11	77.48	**90.32**	76.42	88.32	79.22
7	**100**	79.52	40.41	91.27	73.11	**100**	99.16	86.61	90.26
8	45.18	42.71	45.22	14.38	84.48	55.24	71.34	40.41	**86.39**
9	99.32	11.34	68.17	92.34	75.27	99.13	99.44	99.19	**99.47**
OA	74.11	79.81	68.43	62.28	78.24	82.24	85.38	83.72	**87.74**
AA	78.36	69.20	67.82	70.37	78.36	87.41	86.67	83.77	**89.33**
Kappa	70.25	74.12	63.15	53.26	77.84	77.16	80.10	78.63	**83.99**

TABLE VIII
EXPERIMENTAL RESULTS (%) OF DIFFERENT ALGORITHMS ON SA

Class	HybirdSN	SSRN	A2S2K	SSCL	DFSL	DCFSL	CMFSL	Gia-CFSL	CPSRCL
1	**100**	75.82	**100**	**100**	97.75	**100**	**100**	**100**	99.58
2	**100**	74.99	99.52	99.51	95.21	**100**	98.27	99.24	**100**
3	96.17	**100**	96.35	89.35	99.01	**100**	97.18	94.35	96.63
4	97.33	98.36	83.45	83.43	92.16	92.29	98.33	98.81	**99.96**
5	86.47	53.36	89.40	97.75	97.27	94.11	**98.28**	97.36	90.85
6	**100**	77.81	99.87	95.44	96.35	99.34	99.71	99.82	99.99
7	**100**	**100**	98.55	97.38	98.18	**100**	99.34	**100**	99.85
8	78.52	62.28	89.57	62.45	**92.08**	65.22	63.12	70.74	83.87
9	89.11	99.37	99.86	99.81	97.43	97.19	99.34	**100**	**100**
10	94.68	**96.25**	90.18	94.51	93.29	88.38	95.81	92.16	88.11
11	95.81	55.31	97.11	99.33	90.33	98.21	97.54	95.15	**99.39**
12	91.24	92.24	97.31	79.51	94.72	95.38	94.34	96.35	**98.13**
13	99.38	96.74	96.81	99.67	87.29	99.41	**100**	**100**	98.90
14	60.56	47.66	81.28	96.53	70.24	95.66	**100**	97.82	98.78
15	53.48	65.12	60.65	82.81	68.11	71.84	**85.81**	73.22	75.30
16	**100**	87.52	99.72	91.44	95.37	95.27	94.38	97.43	92.92
OA	83.82	77.68	88.27	86.32	84.88	86.83	89.45	89.24	**91.64**
AA	90.17	80.18	92.48	91.81	91.55	93.27	**95.23**	94.53	95.14
Kappa	81.62	77.48	86.85	84.15	86.81	85.34	88.71	87.59	**90.69**

best classification results, and the classification accuracy of our algorithm for class 11 "Soybean-mintill" is improved by more than 30% compared with other four FSL-based methods, which is the category with the largest number of samples in the IP dataset. This may be because the prototype calibrated using the CSPSR module is more accurate with a larger number of samples. Fig. 11 shows the classification maps, which also clearly shows the superiority of the CPSRCL algorithm.

Table VII shows the results on the UP dataset. The proposed CPSRCL method offers the best overall results for the three indicators of OA, AA, and k coefficients and exceeds the second-best CMFSL by almost 3% in all three metrics. It can be seen that the classification accuracy of the FSL-based algorithms is generally better than that of the DL-based methods in the case of a small number of labeled samples. Fig. 12 shows the classification maps. Specially, it can be seen that our algorithm has better results on classes 4 and 8.

Table VIII lists the experimental results on the SA dataset. We can see that both the FSL-based and deep-learning algorithms can achieve good results due to the regular distribution of samples in the SA dataset. The proposed CPSRCL method still generates the best results. For categories 2, 4, 8, 11, and 12, the proposed CPSRCL yields the best classification results. Compared with Gia-CFSL and DCFSL with the same settings, our method increases OA by 2.4% and 4.81%, AA by 0.61% and 1.87%, and k coefficient by 3.18% and 5.35%, respectively, which validates our proposed modules. Fig. 13 shows the classification results of different algorithms.

Table IX lists the experimental results of different algorithms on the HHK dataset. The proposed CPSRCL algorithm achieves the best results in terms of OA and k coefficients and outperforms the second-best DCFSL algorithm by 3.49% and 3.77%, respectively. It can be seen that our algorithm produces the highest classification accuracy in nine categories, especially in classes 17 and 18.

C. Ablation Analysis

The proposed CPSRCL uses SCL to increase feature discriminant ability, the CSPSR module to refine the support set prototypes, and the LDDA strategy to alleviate the domain shift problem. To verify the effectiveness of these modules,

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TABLE IX
EXPERIMENTAL RESULTS (%) OF DIFFERENT ALGORITHMS ON HHK

Class	HybirdSN	SSRN	A2S2K	SSCL	DFSL	DCFSL	CMFSL	Gia-CFSL	CPSRCL
1	57.58	64.89	**100**	76.65	85.01	85.70	98.24	82.73	88.40
2	95.76	92.23	99.87	96.36	98.87	98.99	98.50	97.98	**100**
3	65.97	56.60	59.48	49.05	56.91	83.81	57.39	**100**	72.38
4	**100**	98.62	**100**	96.39	**100**	92.16	**100**	84.32	71.89
5	89.19	58.10	77.78	81.71	99.95	73.72	**100**	76.92	58.97
6	**100**	70.03	89.49	93.75	92.36	96.52	98.63	94.37	96.36
7	75.63	93.75	98.89	78.76	83.81	98.33	97.73	93.33	**100**
8	80.25	97.50	70.80	**97.98**	74.23	93.85	52.22	95.38	93.85
9	84.19	89.08	86.98	93.93	92.88	70.95	95.89	73.68	**95.27**
10	90.49	90.35	95.63	97.98	92.54	97.17	97.14	99.27	**100**
11	99.57	99.15	**100**	**100**	99.78	**100**	95.63	**100**	**100**
12	91.98	84.58	96.47	81.44	93.03	**100**	95.91	99.44	99.98
13	59.85	**82.17**	78.54	61.38	74.81	74.68	76.24	72.77	63.42
14	99.58	99.20	99.45	98.69	95.95	**100**	99.24	**100**	**100**
15	89.89	99.55	99.51	99.32	98.10	99.67	86.42	97.33	**99.97**
16	48.19	68.75	89.17	37.59	84.33	**93.75**	87.97	85.16	92.19
17	54.46	43.97	46.55	62.21	54.65	68.55	69.04	73.66	**79.57**
18	37.66	47.15	61.54	68.35	49.09	92.06	58.51	80.95	**100**
OA	81.95	81.55	86.74	85.04	87.05	90.17	89.25	90.14	**93.66**
AA	78.90	79.76	86.12	81.75	84.80	**89.99**	86.93	89.29	89.57
K	80.42	80.05	85.66	83.84	85.98	89.33	88.35	89.29	**93.10**

TABLE X
ABLATION ANALYSIS OF DIFFERENT MODULES OF CPSRCL

	SCL	LDDA	CSPSR	OA (%) on each dataset			
				IP	UP	SA	HHK
Methods				67.28	80.22	87.25	86.13
	✓			73.87	84.37	89.32	88.91
	✓	✓		74.63	85.91	90.27	89.62
	✓		✓	76.22	87.69	91.17	91.27
	✓	✓	✓	**77.65**	**87.74**	**91.64**	**93.66**

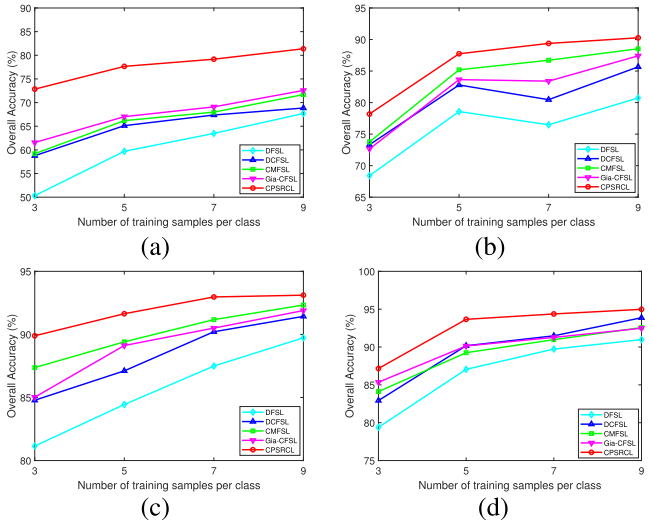


Fig. 14. OA (%) of FSL-based methods with different numbers of labeled samples on four datasets. (a) IP. (b) UP. (c) SA. (d) HHK.

we perform ablation experiments on four datasets and show the corresponding results in Table X.

The first row in Table X is about the classification accuracy without these three modules, which can be considered as the baseline. As can be seen from the table, if we only add SCL to the baseline model, the OA of all four datasets can be improved by about 3%–6%, especially for the IP dataset by 6.61%. It verifies that the SCL module enables the model to extract more discriminative features. Then, we add LDDA and CSPSR to the model, respectively, and the classification accuracy further improves. When all three modules are used, the OA is the highest. It is clear that all three modules are able to enhance the classification performance of the model.

D. Effect of the Labeled Samples

We conducted comparison experiments to verify the impact of different numbers of labeled samples on each algorithm. For the FSL-based methods, three, five, seven, and nine labeled samples per class in the target domain dataset are selected for the training and the rest for testing, and the experimental results are shown in Fig. 14. It is obvious that the OA of five methods improves with increasing labeled samples, and our CPSRCL consistently outperforms other methods. The above results show that our algorithm is able to adapt to changes in the number of labeled samples and may be more advantageous in the case of small-sample classification.

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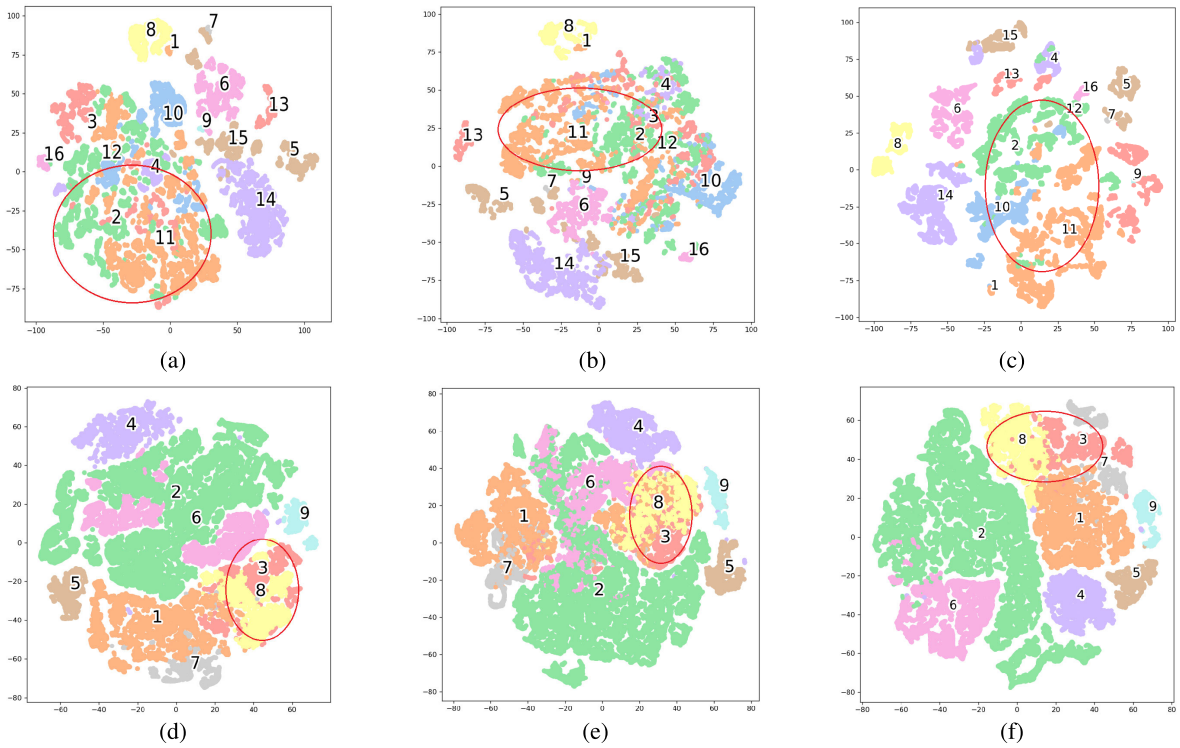


Fig. 15. t-SNE scatter plots on IP and UP datasets. (a) 2-D features visualization of DCFSL on IP. (b) 2-D features visualization of Gia-CFSL on IP. (c) 2-D features visualization of CPSRCL on IP. (d) 2-D features visualization of DCFSL on UP. (e) 2-D features visualization of Gia-CFSL on UP. (f) 2-D features visualization of CPSRCL on UP.

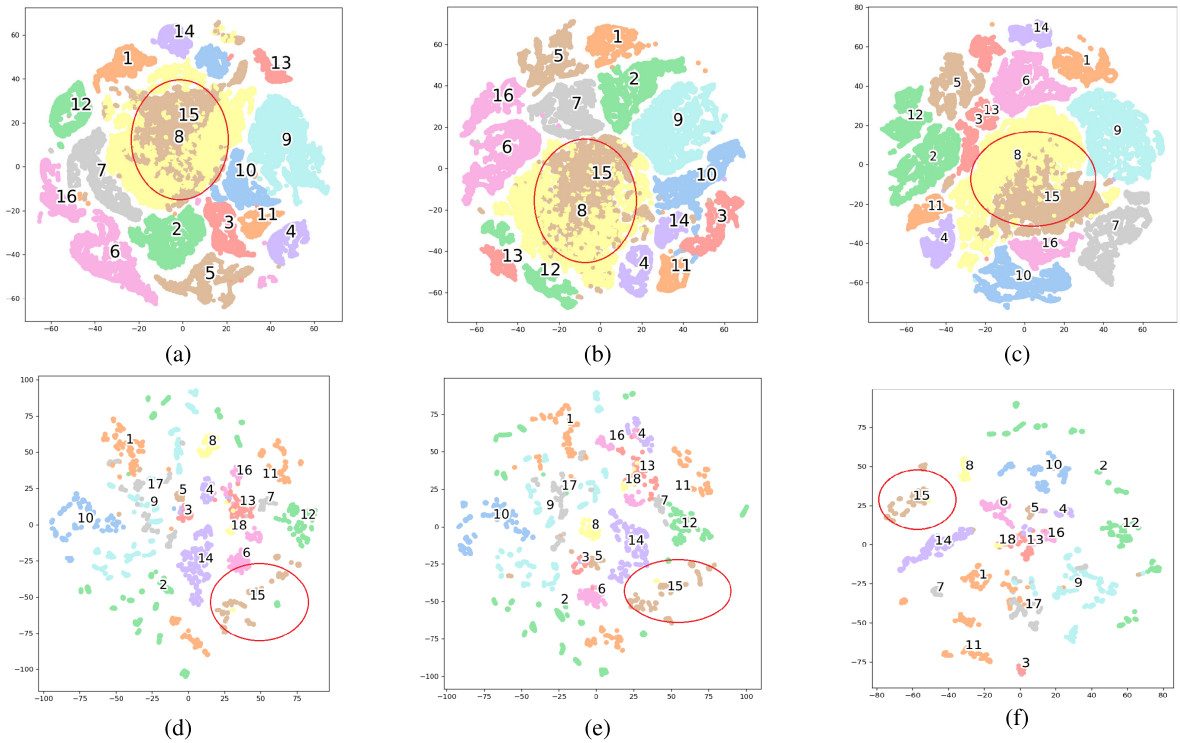


Fig. 16. t-SNE scatter plots on SA and HHK datasets. (a) 2-D features visualization of DCFSL on SA. (b) 2-D features visualization of Gia-CFSL on SA. (c) 2-D features visualization of CPSRCL on SA. (d) 2-D features visualization of DCFSL on HHK. (e) 2-D features visualization of Gia-CFSL on HHK. (f) 2-D features visualization of CPSRCL on HHK.

E. Feature Visualization

To demonstrate the advantages of the proposed CPSRCL method more visually, the t-distributed stochastic neighbor embedding (t-SNE) characteristics of three FSL-based methods (DCFSL, Gia-CFSL, and CPSRCL) are displayed.

Fig. 15 shows the feature distribution of DCFSL, Gia-CFSL, and CPSRCL on the IP and UP datasets, respectively. It can be clearly seen that our CPSRCL is significantly better than the other two algorithms, and the between-class scatter in Fig. 15(c) and (f) is larger. Compared with the other

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TABLE XI
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS

Method		DFSL	DCFSL	CMFSL	Gia-CFSL	OUR
Params(M)		0.03	4.26	0.62	0.31	4.28
Training time (s)	IP	6367	1942	3541	5781	1565
	UP	6367	1696	3954	3497	922
	SA	6367	1949	4217	5992	1729
	HHK	6367	3635	4951	7051	1624

two algorithms, CPSRCL can distinguish many confusing categories efficiently, such as class 2 "Corn-notill" and class 4 "Corn" in IP, and class 3 "Gravel" and class 8 "Bricks" in UP.

On the SA dataset in Fig. 16(a)–(c), all three methods show better results. In particular, our algorithm has better results for class 8 "Grapes untrained" and class 15 "Vinyard untrained," which are difficult to be distinguished. For the HHK dataset in Fig. 16(d)–(f), our algorithm also provides the best overall performance, where features of the same class are clustered closely.

The feature distribution figures show that the proposed CPSRCL is able to extract more discriminative features compared with DCFSL and Gia-CFSL methods. The domain-specific features maintained by the LDDA module, the interclass differences reinforced by the SCL, and the high-quality prototypes refined by the CSPSR module all enhance the learning ability of the model to some extent.

F. Analysis of Computational Complexity

Table XI lists the trainable parameters (M) and training time (S) of five FSL-based methods, which are DFSL, DCFSL, CMFSL, Gia-CFSL, and CPSRCL. It can be clearly seen that our CPSRCL has the largest number of parameters because of more modules. Notwithstanding, our algorithm has a faster computational speed, because the training epoch is much smaller than other algorithms.

V. CONCLUSION

In this article, we have proposed a CPSRCL network and applied the model for small-sample HSIC. A category-specific mapping is designed to update the prototype of support set, allowing the prototypes to be refined adaptively in conjunction with the category information and alleviating the problem of prototype instability caused by noisy samples effectively. To handle both the distribution differences and category differences in source and target domains, an LDDA module is designed to align the global distribution of two domains while maintaining the domain-specific information. In addition, SCL is used to enhance intraclass tightness and interclass dispersion of sample features. The classification results on four public datasets demonstrate the feasibility of the CPSRCL method. Considering that collecting samples by category to construct the target domain also has some difficulties in practical applications, we will further explore the applicability of zero-shot learning for HSIC in our subsequent studies.

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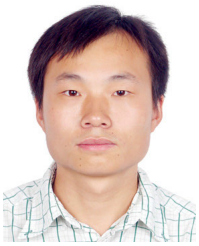
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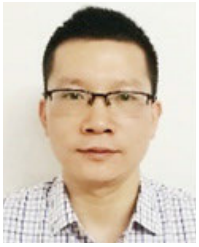
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